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Research article

Effects of La and Y on the microstructure and mechanical properties of $NbMoTiVSi_{0.3}$ refractory high entropy alloys

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ABSTRACT

Effects of La and Y on the phase composition, microstructure and mechanical properties of $NbMoTiVSi_{0.3}$ RHEAs were systematically studied. The results show that all the alloys consist of BCC and M_5Si_3 phases, and most of La and Y distribute at the grain boundary. With the addition of La and Y, the morphology of M_5Si_3 phase changes from large net-like structure to spherical and blocky structure, meanwhile the H, E and H^3/E^2 of BCC and M_5Si_3 phases are improved. The yield strength of RHEAs increases from 1641.10 MPa to 1920.93 MPa for 0.5 at.% La and 1991.59 MPa for 0.5 at.% Y, respectively. Meanwhile, the compression plasticity of RHEAs increases from 12.68 % to 16.82 % for 0.5 at.% La and 16.58 % for 0.5 at.% Y, respectively. The specific strength of the three RHEAs are 236.81, 277.59, 287.80 MPa·cm^3/g, respectively. The addition of La and Y is beneficial to the mechanical properties of $NbMoTiVSi_{0.3}$ RHEA, especially for the Y addition.

Introduction

With the rapid development of gas turbines and aircraft engines, the demand for advanced high temperature materials with excellent mechanical properties is increasing. However, the service temperature of traditional nickel-based superalloy is close to its service temperature limit, which can not meet the service demand of next generation engines and has become a "bottleneck" problem [1]. In recent years, the exploration of refractory high entropy alloys (RHEAs) can effectively solve the serious problem of strength decline of superalloys at high temperature. For example, Senkov et al. [2] prepared NbMoTaW and NbMoTaWV RHEAs with high strength by vacuum arc melting in 2010, which maintained higher strength at high temperature than typical superalloys (such as Inconel 718 and Haynes 230). It is reported that the RHEAs exhibited high strength and hardness, good compression plasticity, excellent wear resistance and corrosion resistance [3]. Due to the introduction of high melting point elements, they also exhibit good high temperature performance. It is expected to replace the traditional nickel-based alloy for application in the aerospace field [4].

RHEAs are generally composed of more than three refractory elements (e.g. Nb, Mo, Ta, Ti, W, V), of which the principal elements contents are 5–35 at.%. However, the RHEAs generally have a disadvantage of high densities (9–14 g/cm^3) [15–17]. For this, adding Si is a common method to reduce the density and regulate comprehensive performance of the RHEAs. When Liu et al. [5] added 14.9 at.% Si into the $HfMo_{0.5}NbTiV_{0.5}$ RHEAs, they found that the density of RHEAs decreased by 5.8 %. Meanwhile, the yield strength of alloy increased from 1260 MPa to 2134 MPa, and the hardness increased from 403 HV to 612 HV. Guo et al. [6] prepared $NbMoTaWSi_x$ RHEAs by spark plasma sintering (SPS), and the density of RHEAs decreased from 13.44 g/cm^3 to 12.23 g/cm^3 as the molar ratio x increased from 0 to 5. Moreover, the yield strength increased from 1217 MPa to 1883 MPa, and the fracture strain increased from 3.8 % to 5.8 %. However, the addition of Si can easily facilitate the formation of silicide in the RHEAs. And the silicide is an intermetallic compound with intrinsic brittleness, which results in the prepared RHEAs exhibiting typical brittle fracture characteristics after compression as a result [6–8].

In general, surface adsorption of active elements (such as K, Ca, Y, etc.) can constrain the orientation growth of intermetallic compounds and then facilitate it spheroidization and refinement, which may promote the strength and toughness of alloy [9]. This is mainly

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because active elements tend to segregate at the front of the solid-liquid interface, which hinders the movement of the solid-liquid interface and improves the morphologies of intermetallic compounds [10]. Many reports have also confirmed that appropriate amount of rare earth elements could refine the microstructure of intermetallic compounds (such as Cr_7C_3, Fe_2B, Fe_3C, $MgSi_2$, AlMg), and improve the plasticity and toughness of alloys as a result [11–14]. Xu et al. [11] studied the influence of different contents of La elements on the $NbMoTiVSi_{0.2}$ RHEAs. They found that, the La enriched at grain boundaries, and the grain size gradually decreased, then the yield strength of alloys gradually increased. When 0.3 wt.% La was added into Mg-1Si alloy by Han et al. [12], eutectic Mg_2Si phase changed from thick long strip to fine dot shape and fiber shape. Bedolla et al. [13] studied the effect of RE-Bi composite modification on the microstructure of carbide. With the increase of modification, the disconnection and spheroidization of carbides in low chromium white cast iron were more obvious. The reduction of size and continuity in the eutectic carbides improved fracture toughness and wear resistance under dry sliding conditions. Yi et al. [14] found that the addition of K would form an adsorption film on the surface of M_2B, thus inhibiting the growth of hard M_2B phase, making the shape of M_2B tend to become spherical or blocky structure, and finally improving the toughness and wear resistance of Fe-2 wt.%B alloy.

Based on these, suitable contents (0.5 at.%) of La and Y were added into $NbMoTiVSi_{0.3}$ RHEAs to improve the mechanical properties. The samples were prepared by vacuum arc melting method. The structural morphology and mechanical properties of the alloys were compared, and the strengthening and toughening mechanism were analyzed, so as to provide reference for the further design of high performance RHEAs.

Experiment

Materials

Bulky or granular Nb, Mo, Ti, V, Si, La, Y materials with purity > 99 % were selected as raw materials. The materials excepting for La and Y were cleaned by ultrasonic to remove the oil stains on the surface after matching, and then placed in a water-cooled copper crucible. Under high purity argon atmosphere, the $NbMoTiVSi_{0.3}$, $(NbMoTiVSi_{0.3})_{99.5}La_{0.5}$, $(NbMoTiVSi_{0.3})_{99.5}Y_{0.5}$ ingots (named H, H-La and H-Y, respectively) were prepared by vacuum arc melting.

High purity Ti was used as the getter of residual gas in the high vacuum chamber, and the working current was 240–280 A. To ensure chemical homogeneity, each ingot was remelted for five to ten times with electromagnetic stirring.

$$\rho_{\text{cal}} = \frac{\sum C_i A_i}{\sum \frac{C_i A_i}{\rho_i}} \tag{1}$$

Where, ρ_i is the density of component i, A_i is the atomic mass of component i, and C_i is the content of component i. The theoretical densities of the three RHEAs are calculated as 6.93, 6.92 and 6.92 g/cm^3, respectively. Compared with other RHEAs [2,5,6,15–21], the densities are greatly reduced, as shown in Fig. 1.

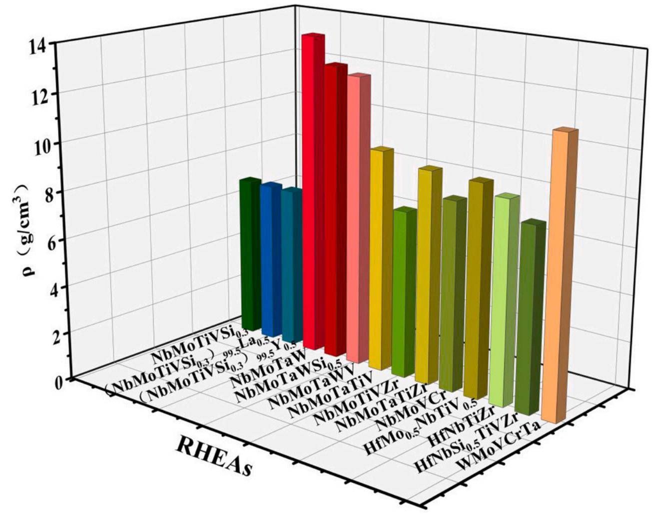

Fig. 1. Comparison among densities of RHEAs [2,5,6,15–21].

Characterization

The samples were cut into 10 × 10 × 8 mm samples by the wire cutting machine. After grinding and polishing, the phase constitution was analyzed on X-ray diffraction (XRD, D/MAX-2500) with Cu-Kα radiation. The XRD was worked at 40 kV and 40 mA with the range of 20°− 90° and the speed of 5°/min. The specimens were etched by Kroll acid (2 % HF + 3 % HNO_3 +95 % C_2H_5OH) for 30 s. The morphologies and chemical composition of the specimens were analyzed by scanning electron microscope (SEM, PHILIP-XL30) equipped with energy dispersion spectroscopy (EDS) instrument. The microstructure constitution was analyzed by using transmission electron microscopy (TEM, FEI Tecnai F30). The TEM samples with the size of ⌀5 × 60 µm were thinned by an ion thinning apparatus (Gatan 691), finally tending to electron transparency. And the entire ion thinning process was divided into three stages: At the first stage, the voltage, incident angle and working time were 5 KeV, 6° and 50 min, respectively; At the second stage, they were 4 KeV, 4° and 50 min, respectively; At the third stage, they were 3 KeV, 3° and 30 min, respectively. After ionic polishing, the samples were analyzed by electron backscatter diffraction (EBSD, AMETEK). The acceleration voltage, step size and working distance of the EBSD samples were 20 KV, 0.5 µm and 15 µm, respectively. The initial data detected by EBSD was analyzed by TSL OIM software.

Based on the GB/T 7314–2005, compressive test was conducted with a Instron universal tester (E45.305) at room temperature, with a strain rate of 1×10^{-3} s^{-1}. The compressive samples were machined into a cylindrical bar with sizes of ⌀4 mm × 6 mm. More than 3 samples were tested for each alloy. The nanoindentation test was performed on the G200 Nano indenter (a Berkovich diamond tip, radius ~200 nm) with a load of 10 mN and a loading rate of 5 mN/min. The displacement (depth) of the indenter was continuously monitored and the load-displacement (*P-h*) curve was drawn. Each nanoindentation test was measured at least 5 times, and then averaged it as final value.

Result and discussion

Microstructure

Fig. 2(a) and (b) show the XRD patterns and local magnification of the three RHEAs. As can be seen from the patterns, the alloys are mainly composed of BCC and M_5Si_3 (M = Nb, Mo, Ti, V) phases. However, as the La and Y are added, the peak value of BCC phase moves slightly to the left. Such as, the diffraction angle corresponding to the diffraction peak (110) decreases from 40.36° to 40.30° for 0.5 at.% La and 40.32° for 0.5 at.% Y, respectively. This indicates that the lattice constant of BCC phase decreases with the addition of La and Y. To confirm this phenomenon, the lattice parameter (*a*) of BCC phase is calculated by prague formula according to Eq.(2) below [22].

$$\sin^2\theta = \frac{\lambda^2}{4a^2}(h^2 + k^2 + l^2)d \tag{2}$$

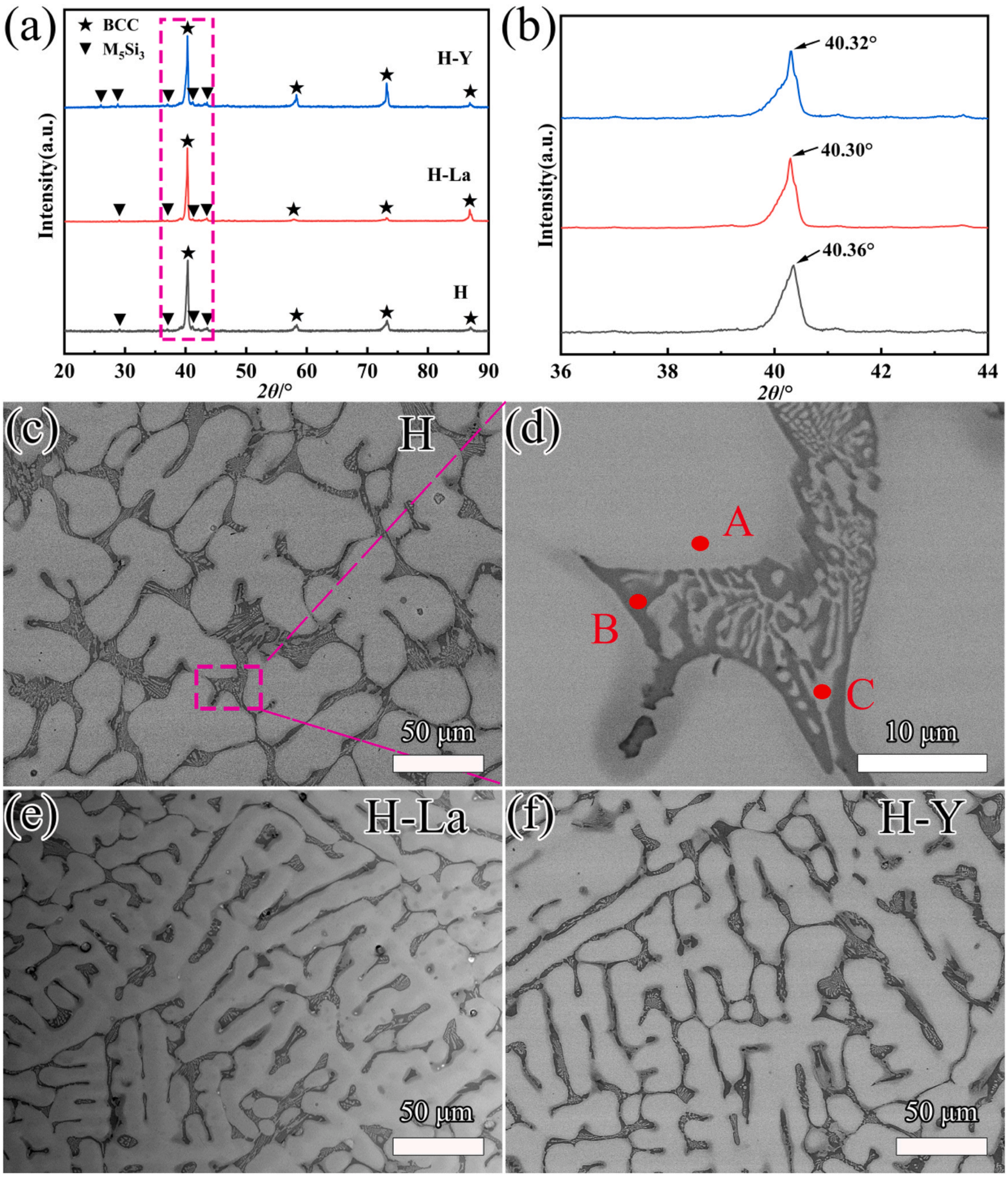

Fig. 2. XRD patterns (a~b) and SEM images (c~f) of the RHEAs.

where θ is the incident angle and λ is the wavelength.

It can be known that a for the BCC phase of H, H-La, and H-Y alloys are 3.0930, 3.0976 and 3.0960 Å, respectively. Such change may be due to the fact that the atomic radius of La (1.879 Å) and Y (1.802 Å) are larger than that of other elements (i.e., Nb, Mo, Ti, V). The calculation results of atomic radius difference parameter (δ) can also prove this deduction as shown in Table 1.

Table 1
VEC, ΔH_{mix}, δ, ΔS_{mix} and Ω of RHEAs.

	H alloy	H-La alloy	H-Y alloy
VEC	4.93	4.92	4.92
ΔH_{mix} (kJ/mol)	-15.67	-15.12	-15.23
δ (%)	5.96	6.47	6.32
ΔS_{mix} (J/mol·K)	12.82	13.02	13.02
Ω	1.95	2.04	2.03

According to the rule of thumb [23,24], five parameters were used to predict phase formation of HEA including the valence electron concentration (VEC), enthalpy of mixing ΔH_{mix} (kJ /mol), entropy of mixing ΔS_{mix} (J/mol·K), atomic radius difference parameter δ (%), and thermodynamic parameter Ω. These parameters can be calculated as following Eqs. (3~7), and the corresponding results are listed in Table 1.

$$VEC = \sum_{i=1}^{n} c_i (VEC)_i \tag{3}$$

$$\Delta S_{mix} = -R \sum c_i \ln c_i \tag{4}$$

$$\Delta H_{mix} = \sum_{i=1, i \neq j}^{n} 4\Delta H_{ij}^{mix} c_i c_j \tag{5}$$

Table 2
Chemical compositions and phase fraction of different microstructures (A, B and C regions) in the three RHEAs.

Alloy	Phase	Phase fraction (%)	Nb (at.%)	Mo (at.%)	Ti (at.%)	V (at.%)	Si (at.%)	La (at.%)	Y (at.%)
H	BCC phase	83.39	26.16	32.14	18.35	19.64	3.72	–	–
	Silicide	12.51	19.12	7.23	31.44	15.12	27.08	–	–
	secondary BCC phase	4.10	22.65	16.15	25.76	22.91	12.53	–	–
H-La	BCC phase	84.07	24.85	32.00	17.38	19.39	3.54	2.84	–
	Silicide	12.04	18.83	5.37	30.31	9.74	32.21	3.54	–
	secondary BCC phase	3.89	21.04	14.76	25.58	24.58	10.43	3.82	–
H-Y	BCC phase	83.31	25.10	32.44	18.00	19.45	3.45	–	1.56
	Silicide	12.94	18.52	4.84	33.20	10.23	32.19	–	1.03
	secondary BCC phase	3.75	20.74	13.41	27.76	23.55	13.42	–	1.13

$$\Omega = \frac{T_m \Delta S_{mix}}{|\Delta H_{mix}|} \tag{6}$$

$$\delta = \sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{r_i}{\bar{r}}\right)^2},\ \bar{r} = \sum_{i=1}^{n} c_i r_i \tag{7}$$

Yang and Zhang [23] proposed that stable high-entropy solid solutions could be formed when $\Omega \geq 1.1$ and $\delta \leq 6.6$. Guo et al. [24] proposed VEC to predict the phase stability of HEAs solid solutions. When VEC ≥ 8, HEAs tended to form FCC solid solutions. While VEC≤ 6.87, it is more likely to form BCC solid solutions. The calculated VEC in this work are 4.93, 4.92 and 4.92 respectively, all of which are lower than 6.87, well corresponding to the VEC judgment method. Senkov et al. [25] proposed that the range of solid solution (SS) and intermetallic (IM) mixing phase was $1.1 \leq \Omega \leq 10$ and $3.6 \leq \delta \leq 6.6$. For the H, H-La, and H-Y alloys, the corresponding calculated values of Ω are 1.95, 2.04, 2.03, and the δ are 5.96, 6.47, 6.32, respectively. It is within the predicted range of SS and IM mixing phases. In conclusion, the phases BCC and silicide obtained in this paper are consistent with the predicted results of the formulas.

From Fig. 2(c)~(f), it can be seen that all the alloys have typical dendrite and interdendrite structures. It is well known that lighter elements have darker contrast in the BSE image, so the dark lamellar structure can be inferred to be the M_5Si_3 phase. Then, the dendritic trunk and arms are BCC phase. The XRD results show that there are only BCC phase and M_5Si_3 phase in the alloys. Therefore, one phase of the interdendrite is the same as the phase in the dendrite, that is, the secondary BCC phase. Due to the complex composition, the RHEAs system is considered as a pseudo-binary M-Si system (M=Nb, Mo, Ti, V). According to the Ti-Si binary phase diagram and Mo-Si binary phase diagram [57,58], the solidification process of M-Si alloy can be summarized as follows. The primary BCC phase firstly separates from the melt, and then the M rich solid solution forms gradually with the decrement of temperature. When it reaches the eutectic temperature, the eutectic reaction occurs, namely the BCC1 phase transforms into the eutectic phases (BCC2 +M_5Si_3). Therefore, the secondary BCC phase actually refers to the eutectic BCC phase [59]. Point A, B and C in Fig. 2(d) are scanned by EDS at high magnification, in which A is primary BCC phase, B is silicide and C is secondary BCC phase. The EDS results are shown in Table 2. It can be seen that Nb, Mo and V are enriched in dendrites (primary BCC phase), while Ti and Si are enriched in the interdendrite region (M_5Si_3 phase). Also, the contents of Ti, Si and V elements in the secondary BCC phase are higher than those in the primary BCC phase, while Nb and Mo elements are less. In conclusion, the difference between the primary BCC phase and the secondary BCC phase is the formation process, formation temperature and chemical composition.

The factor affecting element enrichment can be explained by using ΔH_{mix}, which indicates the bonding strength of component elements. As shown in Table 3, the ΔH_{mix} between Nb and Mo is − 6 kJ /mol, the ΔH_{mix} between Nb and V is − 1 kJ /mol, and the ΔH_{mix} between Nb and Ti is 2 kJ /mol. According to Miedema model [26], this means that Nb, Mo and V tend to be enriched together rather than separation from each other. The ΔH_{mix} value between Si element and metal elements are negative, indicating that the silicide can be formed in the RHEAs. Compared with the ΔH_{mix} of Si element and all metal elements, the ΔH_{mix} of Ti-Si is the most negative, showing that Ti and Si are most inclined to gather together and then from into M_5Si_3.

Table 3
ΔH_{mix} of different atomic pairs.

ΔHmix (KJ/mol)	Nb	Mo	Ti	V	Si	La	Y
Nb	–	-6	2	-1	-56	36	30
Mo		–	-4	0	-35	31	24
Ti			–	-2	-66	20	15
V				–	-48	22	17
Si					–	-73	-73
La						–	0
Y							–

It can be seen from the SEM micrographs (Fig. 2(c)~(f)) that, with the addition of La and Y, the M_5Si_3 phase occurs disconnection and spherodization to a certain extent, and its particle size tends to decrease. In order to analyze the effect of La and Y on the morphology of silicides, the shape factor (K) and grain factor (D) of M_5Si_3 were quantitatively characterized. 20 SEM images (1000x magnification) were colored using Photoshop (as shown in Fig. 3(a)). Subsequently, the perimeter and area of the colored silicides were calibrated by using Image Pro Plus software, and then the K and D values were calculated. The calculation formulas are as follows [27,28]:

$$K = \frac{4\pi A}{L^2} \tag{8}$$

$$D = 2\sqrt{\frac{A}{\pi}} \tag{8}$$

Where A is area; L is perimeter. The value of K ranges from 0 to 1, and the closer the K value is to 1, the more spherical the silicide tends to be. The smaller the D value is, the smaller the particle size of the silicide is. Fig. 3(b) shows the calculated results of K and D values of silicides in the RHEAs. Compared to the H alloy, the K value of the H-La alloy increases from 0.26 to 0.42 with an increment of 61.54 %, while that of the H-Y alloy is improved by 46.15 %. Moreover, the D value decreases from 15.17 (H alloy) to 8.21 for the H-La alloy and 9.03 for the H-Y alloy, respectively. This reveals that the M_5Si_3 of RHEAs tends to spherification and refinement after the addition of La and Y, and the La addition has a better effect on the refinement of silicide than the addition of Y. The phase fraction of BCC (primary and secondary) and silicide phases is also calculated by Image pro plus software, and the results are concluded in Table 2. After adding La and Y, the phase fraction of each phase changed little. The phase fraction of primary BCC phase in the H, H-La, H-Y alloys is 83.39 %, 84.07 % and 83.31 %, respectively, while that of M_5Si_3 phase is 12.51 %, 12.04 % and 12.94 %, respectively. Among the three phases, the phase fraction of secondary BCC phase is the smallest, which is

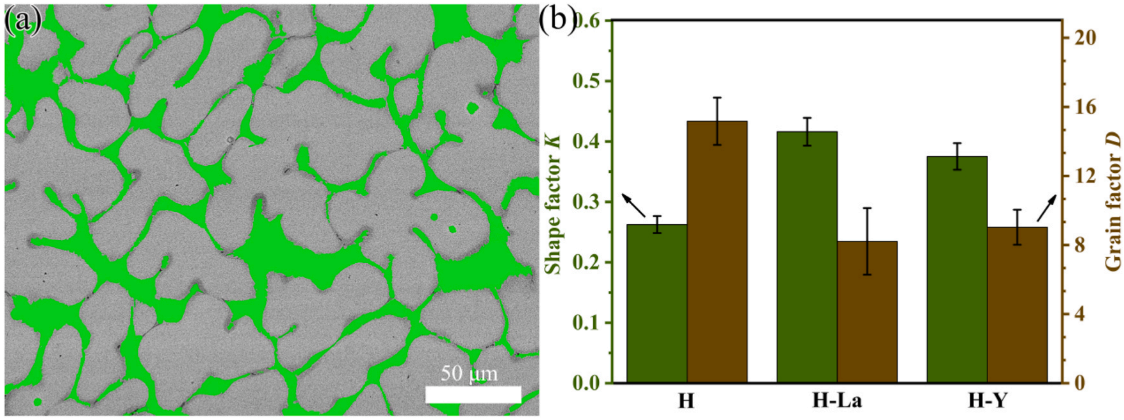

Fig. 3. (a) Schematic diagram of statistics; (b) shape factor *K* and grain factor *D* of silicides.

4.10 % (H alloy), 3.89 % (H-La alloy), and 3.75 % (H-Y alloy), respectively.

Generally, surface active elements (such as K, La, or Ca) tend to segregate on the surface of eutectic phases (such as carbides, borides, or silicides) due to their low interfacial energy, thus inhibiting the growth of eutectic phases [29,30]. In order to verify this phenomenon, the distributions of La and Y in the H-La and H-Y alloys were analyzed by line scanning, and the results are shown in Fig. 4(b) and (c) respectively. From the intensity of elements, it can be seen that La and Y are obviously enriched at the boundary between the BCC and M_5Si_3. This indicates that La and Y may produce composition supercooling and form an adsorption film at the solid-liquid interface of M_5Si_3, thus limiting the growth of M_5Si_3 [31]. Moreover, the film can also inhibit the entry of Nb, Mo and V atoms into the M_5Si_3 phase, which is also confirmed by the spot scan results in Table 2. That is, with the addition of La and Y, the contents of Nb, Mo and V of the M_5Si_3 phase decrease. Therefore, the growth rate of M_5Si_3 is limited by the adsorption film, and M_5Si_3 tends to form spherical or block-shape structure (seen in Fig. 2).

EBSD technique is mainly used for orientation and phase structure determination. Therefore, EBSD detection was carried out for the H, H-La and H-Y RHEAs, and the phase distribution, grain size, Schmidt factor, Kernel average misorientation (KAM) of RHEAs were analyzed. Fig. 5(a) shows the detected grains of RHEAs. According to

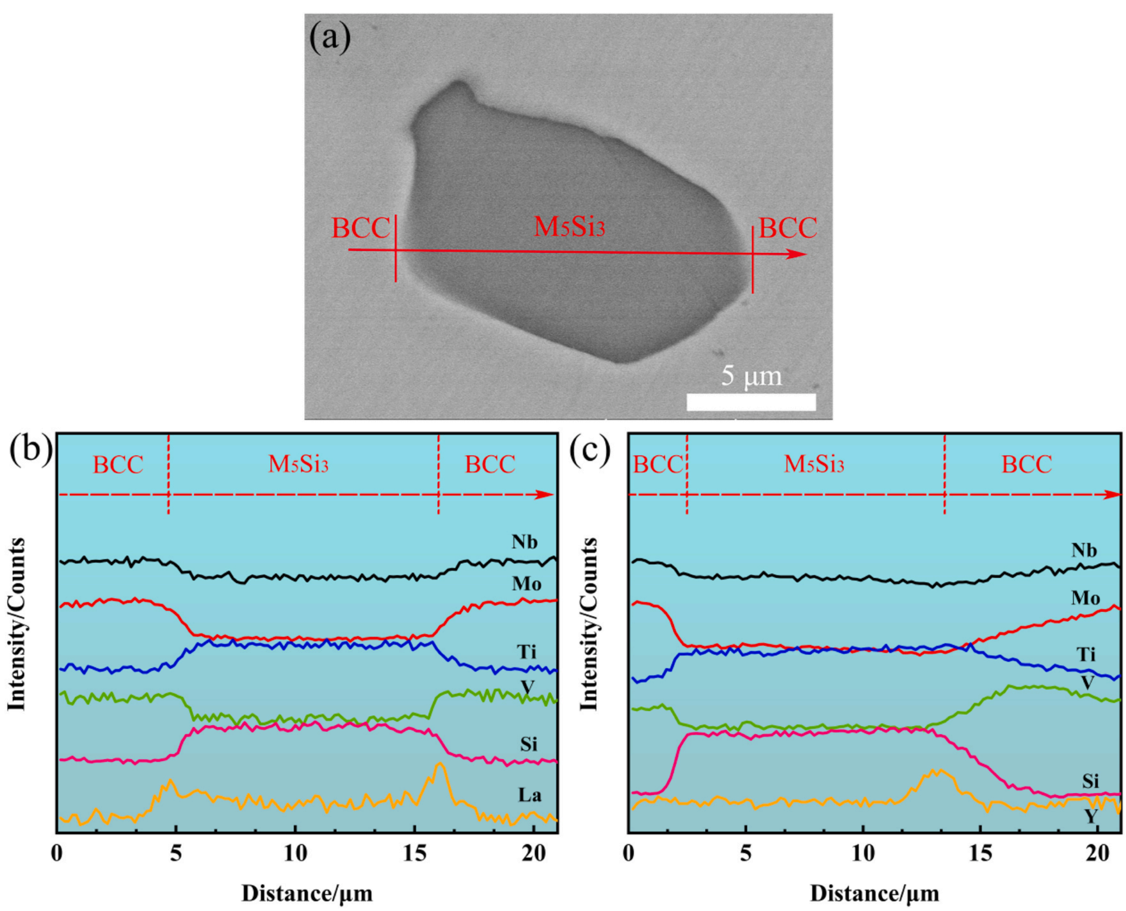

Fig. 4. The distribution of La and Y in the RHEAs. (a) SEM morphology; (b)~(c) linear scanning result: (b) H-La; (c) H-Y.

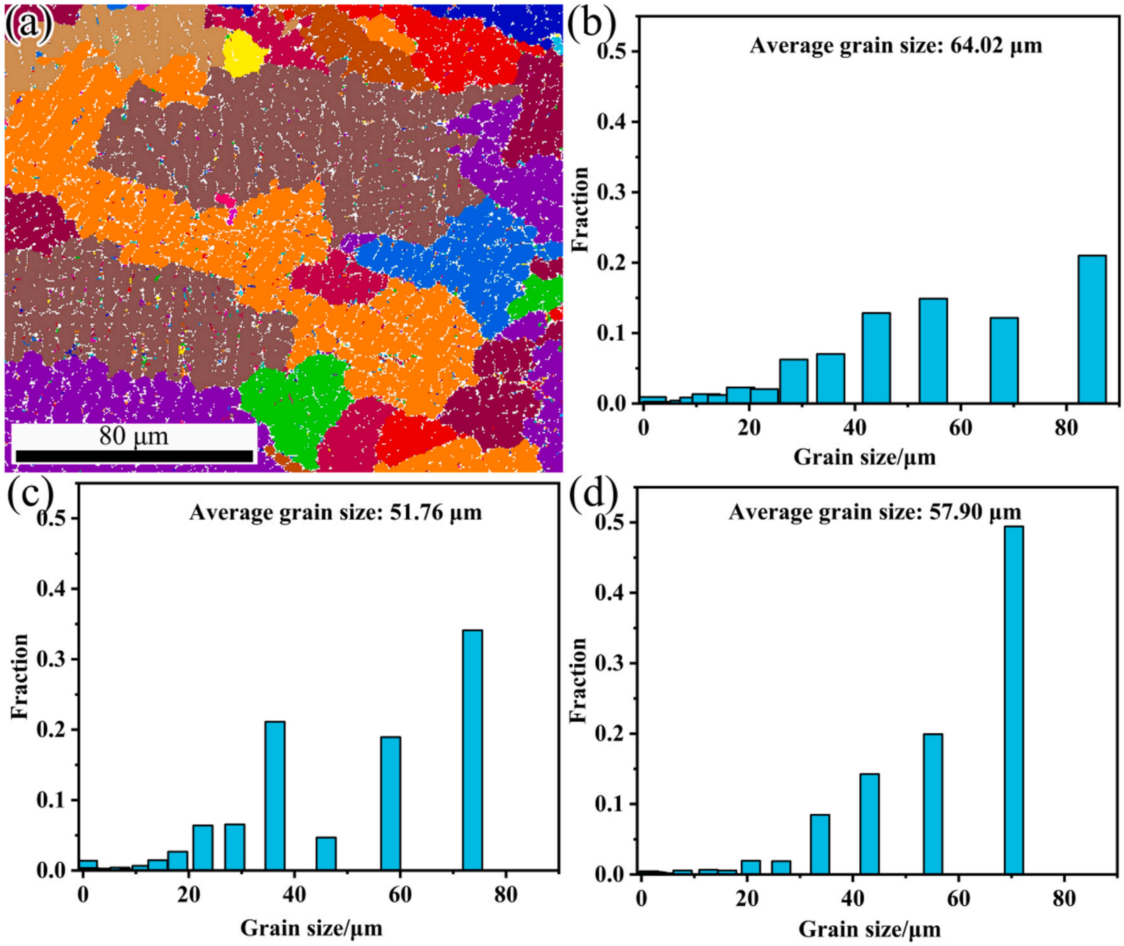

Fig. 5. (a) grain size map of H; (b)~(d) gain size: (b) H; (c) H-La; (d) H-Y.

the grain size diagrams (Fig. 5(b)~(d)), the average grain size of RHEAs decreases from 64.02 to 51.76 and 57.90 μm by adding 0.5 at.% La and 0.5 at.% Y, respectively. According to the classical nucleation theory [32,33], the addition of rare earth elements reduces the surface tension of phase interface and the energy fluctuations of nucleation. Therefore, the addition of La and Y can increase the nucleation rate, preventing crystal growth and grain boundary movement, and then refining the grains of RHEAs.

Fig. 6(a1)~(c1) show the phase maps in the RHEAs obtained by EBSD, displaying the dual phase microstructure, where the red and green regions represent the BCC phase and the M_5Si_3 phase, respectively. Analyzed by OIM software, the contents of BCC phase in the H, H-La, H-Y alloys are 90.20 %, 90.20 % and 90.00 %, respectively, while that of M_5Si_3 phase are 9.80 %, 9.80 %, 10.00 %, respectively. This indicates that the addition of La and Y has no obvious effect on the phase content distribution. Fig. 6(a2)~(c2) show the inverse pole figures (IPFs) maps of the H, H-La and H-Y alloys, respectively. A stereographic unit triangle for crystal directions with assigning red, green, and blue to the 001, 011 and 111 corners respectively for the BCC phase, while the M_5Si_3 phase correspond to 0001, 10–10 and 2–1–10 corners respectively. Fig. 6(a2) shows that the H alloy without adding La or Y exhibits a strong (112) orientation. And orientation increases with rare earth elements addition. That is, the alloy without adding La and Y shows a strong trend of preferred orientation. While the alloys with adding La and Y showed a trend of anisotropic growth. The addition of La and Y can change the growth mode of the alloy to a certain extent, which is beneficial to improve the morphology of the alloys. Fig. 6(a3)~(c3) depict the angle distribution of grain boundaries in the RHEAs. 2°− 15° is the low angle grain boundary (LAGB) marked in red, while 15°− 180° is the high angle grain boundary (HAGB) marked in blue. According to statistics, the fraction of LAGBs increases from 18.4 % (H alloy) to 25.8 % (H-La alloy) and 23.3 % (H-Y alloy). More LAGBs mean higher grain boundary resistance, which means larger work hardening property of RHEAs [34].

The influence of La and Y on the misorientation angle distribution of RHEAs is shown in Fig. 6(a4)~(c4). The average grain boundary misorientation angles of the H, H-La and H-Y alloys can be calculated as 34.32°, 25.33° and 25.17°, respectively. Moreover, the misorientation angles within the range of 0–10° increase significantly with the addition of La and Y. Studies have shown that the increment of small misorientation angles (e.g., 0°~10°) can increase the energy barrier of dislocation penetrating through grain boundary, then inhibiting dislocation slip and improving the strength of alloy [35]. In addition, according to the Voce Hardening Scheme [36], the contribution of dislocation-grain boundary interactions on yield strength and strain hardening of alloy or metal is closely related to misorientation angle. As the small misorientation angle increases, it becomes more difficult for the dislocation to penetrate the adjacent grains, and the stress generated by the dislocation-grain boundary interaction increases.

Fig. 7(a)~(c) display the KAM maps of the H, H-La and H-Y alloys. KAM represents the mean orientation difference between adjacent points. The redder the region shows, the larger the KAM value is. It can be seen that after the addition of La and Y, the green and red regions in the maps increase, especially at the grain boundary area,

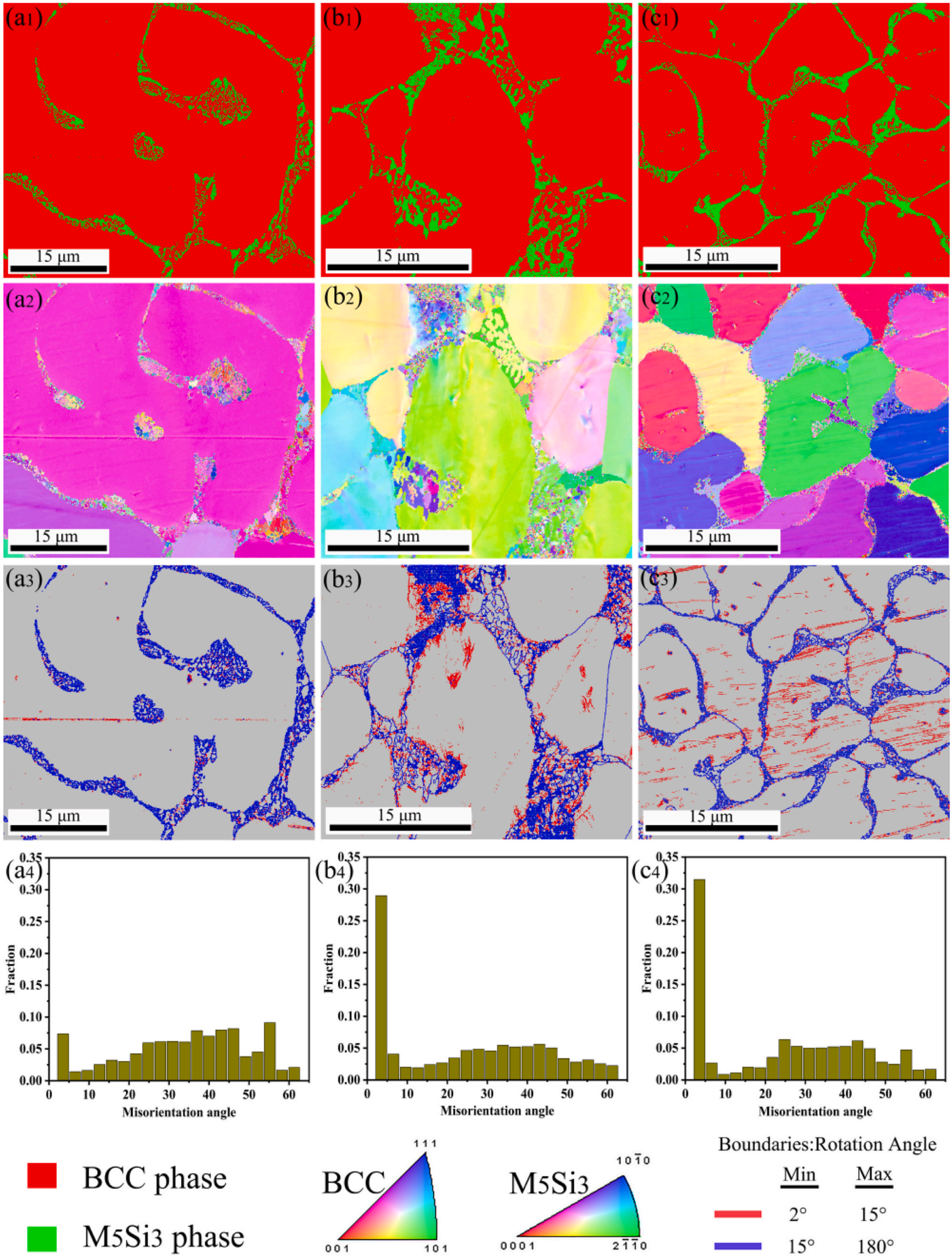

Fig. 6. EBSD dataset of RHEAs: (a) H alloy; (b) H-La alloy; (c)H-Y alloy; (Symbols 1, 2, 3 and 4 show the phase map, IPF map, grain boundary map and misorientation distribution, respectively).

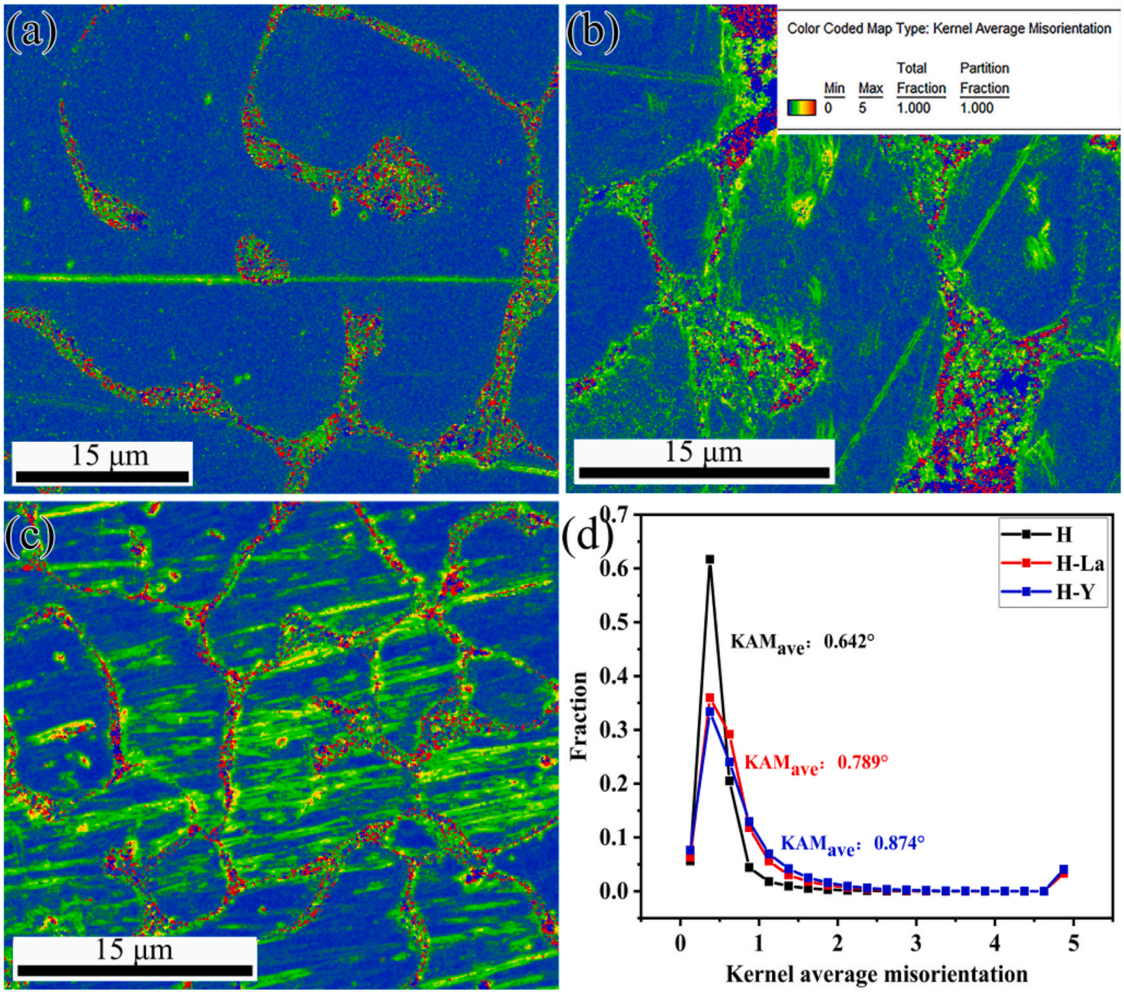

Fig. 7. (a)~(c) KAM maps of RHEAs: (a) H; (b) H-La; (c) H-Y; (d) KAM distribution.

that is dislocation density increases, which means the energy stored at the grain boundary and the local dislocation density increase [37].

This is mainly due to the decrease of grain size caused by the addition of La and Y (Fig. 4). The role of grain boundaries in the inhibition of dislocation propagation becomes more and more obvious, leading to the accumulation of dislocation at the grain boundary. Fig. 7(d) shows the KAM values of the three RHEAs. After adding La and Y, the average KAM value increases from 0.642 to 0.789 and 0.874, respectively, indicating that slightly different orientation differences occur in many areas. This also indicates that the addition of La and Y tends to increase the dislocation density [38].

Schmid factor is an important index to measure the deformation ability of materials under external force, and it strongly affects dislocation slip and twinning deformation. The value of Schmid factor is directly related to the intensity of red color, which means that the redder the color of grains displays, the larger the Schmid factor is [39]. The Schmid factor distribution of RHEAs is displayed in Fig. 8(a)~(c), and the quantitative statistic of Schmid factor is displayed in Fig. 8(d)~(f). It can be clearly observed that each alloy has a large Schmid factor. Gussev et al. [40] proposed that the Schmid factor of soft particle was greater than 0.4, while that of hard particle was less than 0.35. The larger the Schmid factor, the easier the grain deformation. The average Schmidt factors of the H, H-La and H-Y alloys are 0.452, 0.488 and 0.444 respectively, indicating that most grains are still in soft orientation. This shows that the Schmidt factors of RHEAs prepared in this work are relatively high, indicating that plastic deformation is mainly achieved through slip. Large grain Schmid factors may contribute to grain boundary migration [41]. For grains with high Schmid factor, the dislocation is easily activated at grain boundary (seen in Fig. 7). This facilitates the formation of pure shear, leading to the grains occurring in yield easily without rotation or grain reorientation [42].

The bright-field TEM microstructures of the RHEAs have been studied, and the crystal structures of the phase have been determined by selected area electron diffraction (SAED) patterns, as shown in Fig. 9. As can be seen in Fig. 9(b)~(c), a small number of dislocations are formed in the BCC phase after the addition of La and Y (as indicated by the red arrows), which is well consistent with the result of Fig. 7. The lattice distortion caused by the large atomic radius difference of rare earth elements limits the grain growth. Thus there is no enough time for the grain to consume its own dislocations to nucleate, and it is difficult to release the stored energy [60]. The dislocation tangles and accumulates near the silicide, which increases the dislocation density and is beneficial to improve work hardening ability of the alloy, as also reported in Ref.[43].

Fig. 9(d) is the SAED pattern of the region A in Fig. 9(a), which is the BCC phase with lattice parameters of $a = b = c = 3.1314$ Å. Fig. 9(e) shows the SAED pattern of interdendritic silicide (region B) in Fig. 9(a), which shows HCP structure with lattice parameter of $a = b = 3.1425$ Å, $c = 5.2136$ Å. For the H-La alloy, the SAED pattern of the BCC and M_5Si_3 phases are taken in the [111] zone of the BCC

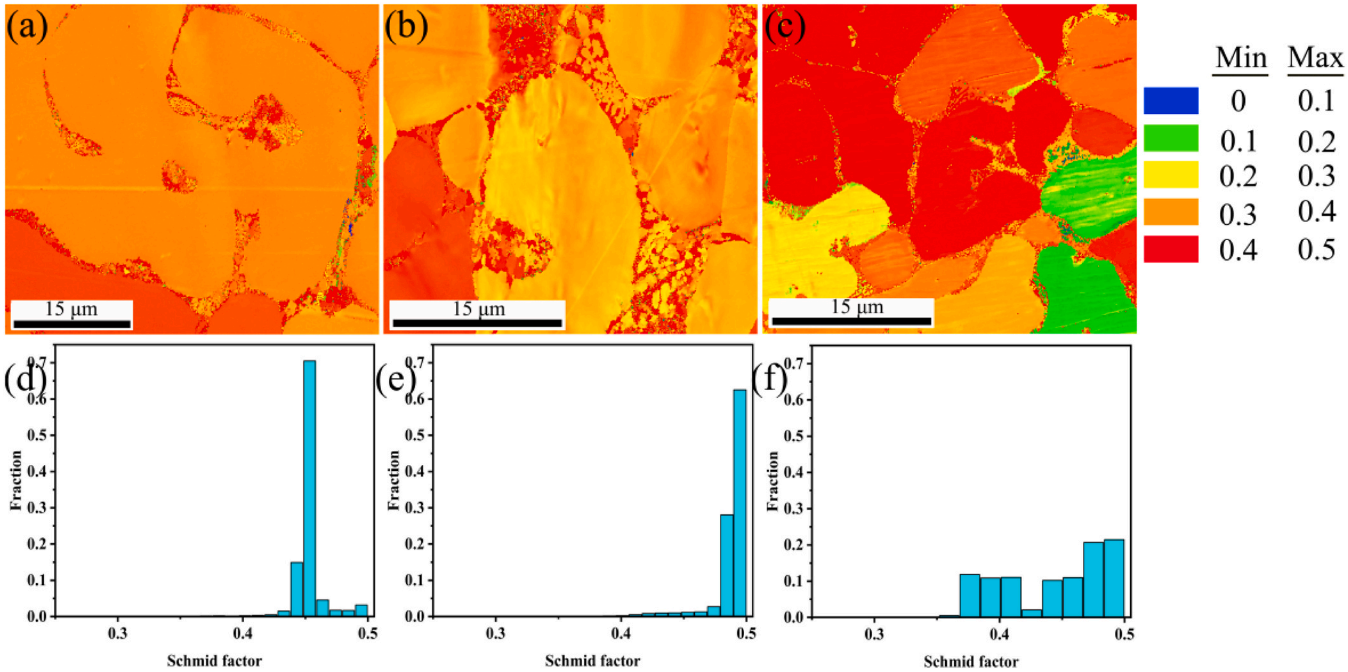

Fig. 8. Schmid factor maps and values of the RHEAs: (a) and (d) H alloy; (b) and (e) H-La alloy; (c) and (f) H-Y alloy.

structure and [1−15] zone of the HCP structure as shown in Fig. 9(f) and (g), respectively. For the H-Y alloy, the SAED pattern of the BCC and M_5Si_3 phases are taken in the [001] zone of the BCC structure and [2−1−10] zone of the HCP structure as shown in Fig. 9(h) and (i), respectively. Here, the lattice parameters of BCC structure in the H-La and H-Y alloys are $a = b = c = 3.1338$ Å, $a = b = c = 3.1339$ Å, respectively. And the lattice parameters of HCP structure in the H-La and H-Y alloys are $a = b = 3.2300$ Å, $c = 5.2744$ Å, $a = b = 3.8391$ Å, $c = 6.2692$ Å, respectively. The SAED patterns reveal that the addition of La and Y has no effect on the phase structures of RHEAs, while decreasing the lattice parameters, which is consistent with XRD results.

Mechanical properties

Nano-indentation

Nanoindentation can be used to accurately measure the nano-mechanical properties of alloy. It can also be used to continuously measure load-displacement changes at nanoscale. The nanoindentation test has been performed on BCC phase and M_5Si_3 phase at 10 mN, and the *P-h* curves are plotted in Fig. 10. The corresponding nanohardness (*H*) and elastic modulus (*E*) values are obtained from the curves and shown in Table 4. For the H alloy, the *H* and *E* of BCC phase are 13.03 GPa and 218.93 GPa, respectively, while that of M_5Si_3 phase are 19.93 GPa and 239.86 GPa, respectively. For

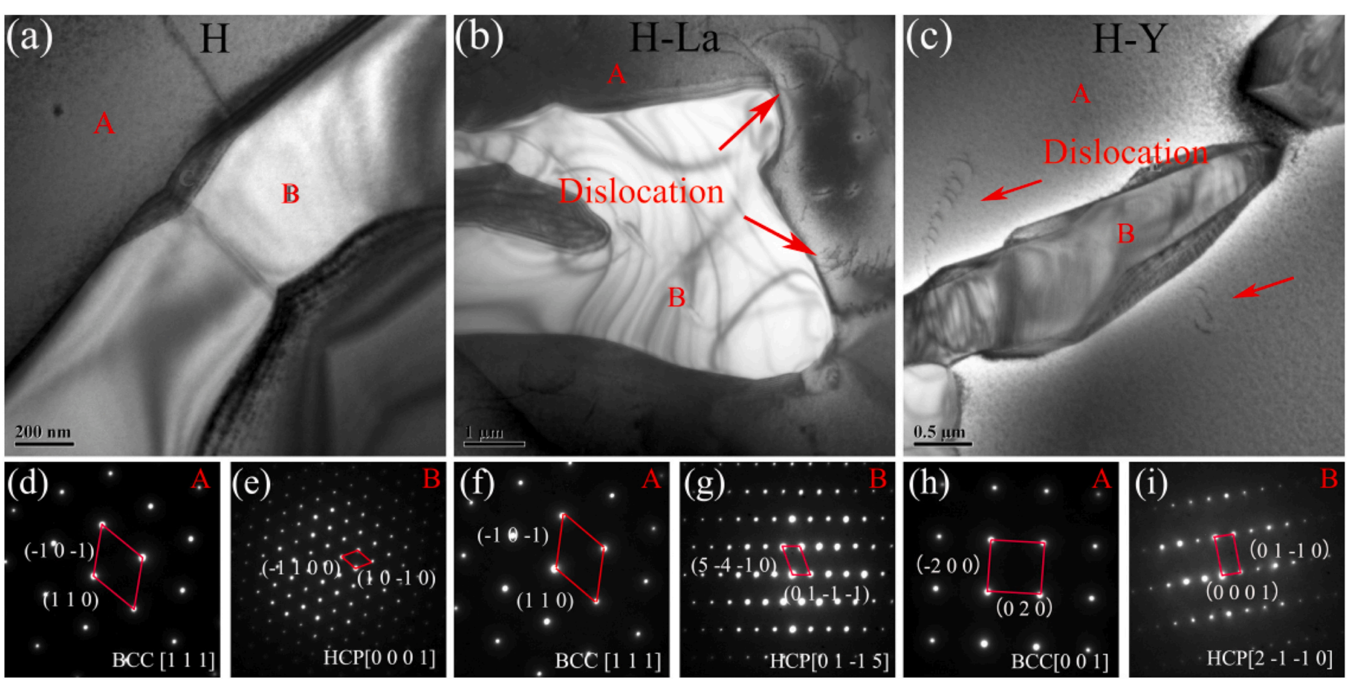

Fig. 9. (a)~(c): bright field TEM micrographs of RHEAs; (d)~(i): SAED patterns.

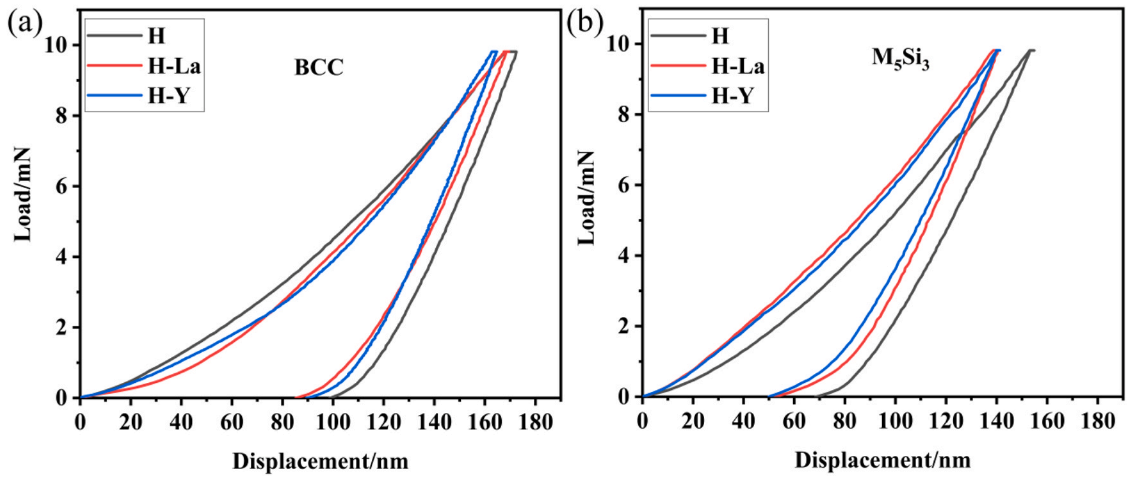

Fig. 10. Load (P) and depth (h) curves of the three RHEAs: (a) BCC phase; (b) M_5Si_3 phase.

Table 4
Nanohardness (H) and elastic modulus (E) of BCC phase and M_5Si_3 phase.

Alloy	Phase	H (GPa)	E (GPa)	H^3/E^2 (GPa)
H	BCC	13.03 ± 0.91	218.93 ± 4.70	0.047 ± 0.008
	M_5Si_3	19.93 ± 2.36	239.86 ± 13.93	0.146 ± 0.041
H-La	BCC	14.39 ± 0.79	225.10 ± 8.49	0.065 ± 0.007
	M_5Si_3	21.07 ± 2.24	254.13 ± 16.36	0.165 ± 0.047
H-Y	BCC	14.78 ± 1.73	229.99 ± 8.50	0.067 ± 0.016
	M_5Si_3	22.14 ± 2.21	267.39 ± 10.06	0.162 ± 0.039

Table 5
Compressive mechanical properties of RHEAs at room temperature.

Alloys	Yeild strength (MPa)	Maximum strength (MPa)	Facture strain (%)	Specific yeild strength (MPa·cm^3/g)
H	1641.10	1927.20	12.68	236.81
H-La	1920.93	2161.61	16.82	277.59
H-Y	1991.59	2166.79	16.58	287.80

the H-La alloy, the H and E of BCC phase increase to 14.39 GPa and 225.10 GPa, respectively, while that of M_5Si_3 phase increase to 21.07 GPa and 254.13 GPa, respectively. With the addition of Y, the H and E of BCC phase increase to 14.78 GPa and 229.99 GPa, respectively, while that of M_5Si_3 phase are 22.14 GPa and 267.39 GPa, respectively. With the addition of La and Y, the nanohardness and elastic modulus of BCC phase and M_5Si_3 phase increase. And the nanohardness of M_5Si_3 phase improves more obviously. M_5Si_3 is a brittle phase with higher hardness and is more prone to generate cracks than BCC matrix.

Another parameter H^3/E^2, called yield resistance, can effectively describe the toughness of a given material [44,45]. The ratio H^3/E^2 of RHEAs was calculated as also shown in Table 4. The H^3/E^2 of the BCC and M_5Si_3 phases increase with the addition of La and Y. It can be concluded that the H-La and H-Y alloys have greater resistance to plastic deformation of surface contact than the H alloy.

Compression

Fig. 11(a) shows the compressive stress-strain curves of the three RHEAs. Specific yield strength is the ratio of yield strength to material density. As an important index of structural materials, the higher the specific yield strength means, the lower the dead weight and the higher the strength of the structure in the application scenario.

It can be seen that the strength and plasticity of RHEAs are improved after the addition of La and Y, and the results are recorded in Table 5. With the addition of La, the yield strength, compression plasticity, and specific strength of RHEAs increases from 1641.10 MPa to 1920.93 MPa, 12.68 % to 16.82 %, and 236.81 MPa·cm^3/g to 277.59 MPa·cm^3/g, respectively. With the addition of Y, the yeild strength, compression plasticity, specific strength increases to 1991.59 MPa, 16.58 %, 287.80 MPa·cm^3/g, respectively. In Xu's work [11], the yield strength and fracture strain of $(NbMoTiVSi_{0.2})_{99.5}La_{0.5}$ RHEA were 1929 MPa and 15.28 %, respectively. The improvement of

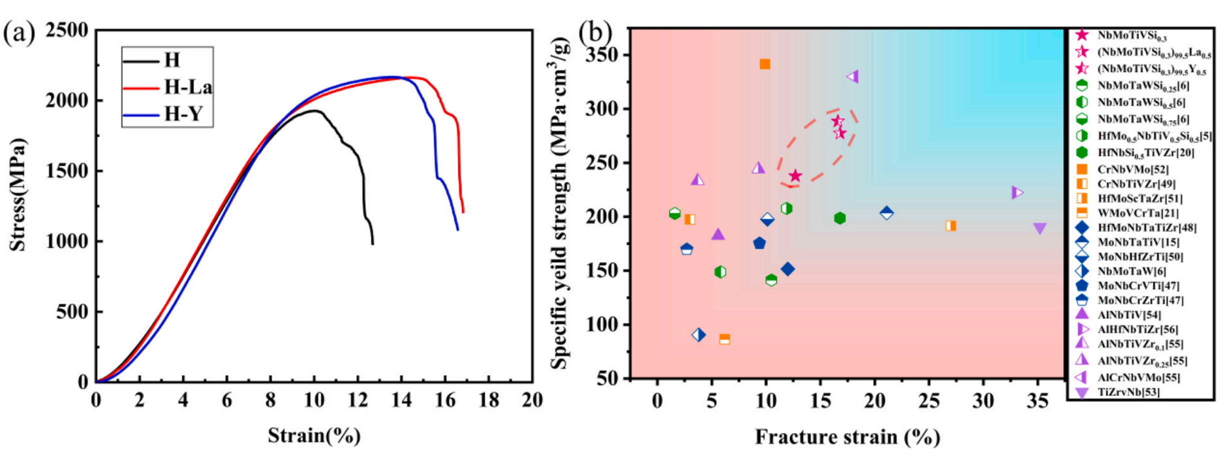

Fig. 11. (a) Compressive stress-strain curves of RHEAs; (b) Comparison among specific yield strength of the RHEAs in present work and other RHEAs.

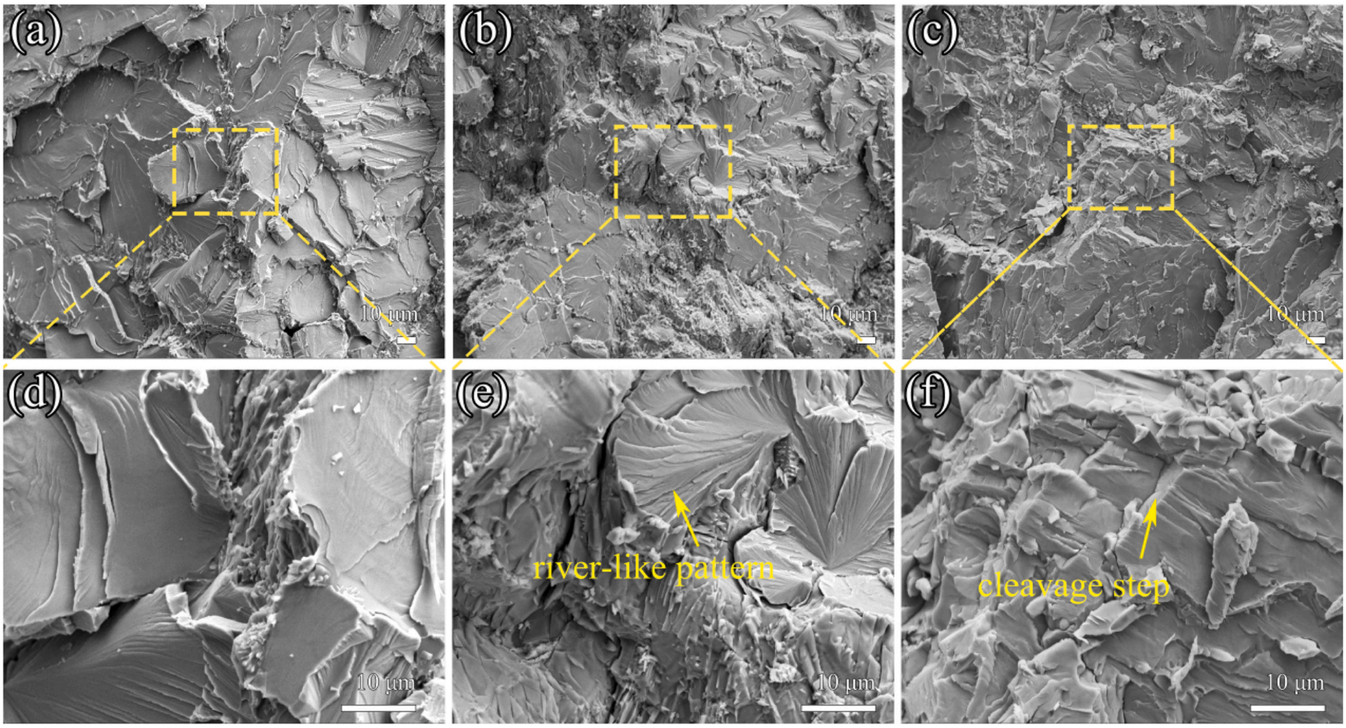

Fig. 12. Compressive fracture morphologies of RHEAs: (a) and (d) H alloy; (b) and (e) H-La alloy; (c) and (f) H-Y alloy.

plasticity is because the morphology of silicides tends to disconnection, spheroidization and refinement after the addition of La and Y (seen in Fig. 3). The brittle net-like silicide is prone to crack propagation, while the soft BCC phase is more likely to absorb the energy of crack propagation. Therefore, the spherodized and discontinuous silicide can effectively prevent crack growth and improve the ductility of alloy. As described by Xu et al. [46], after hot rolling $TiVNbTaSi_{0.1}$ RHEAs at 1200 °C, they found that the eutectic net-like silicide was refined and changed into strips along the rolling direction, and then the plasticity increased from 1.4 % to 4.2 %.

The improvement of strength is mainly due to the following reasons: 1. The addition of La and Y induces grain refinement (as shown in Fig. 5), which plays a role of fine grain strengthening; 2. The solid solution of La and Y in the BCC phase (Table 3), resulting in the lattice distortion (Fig. 2(a)) and facilitating the formation of dislocation (Fig. 9), plays a certain effect of solid solution strengthening and dislocation strengthening. 3. A small amount of La and Y dissolves into the M_5Si_3 phase (seen in Table 2), improving its H and H^3/E^2 (Table 4), and then promoting the strength of alloy.

At present, structural materials tend to be lightweight, especially for aeronautic materials, which have high demand for lightweight and high-strength materials. As mentioned before, RHEAs generally have high density. However, too high density limits the application of RHEAs in aerospace, marine shipbuilding, automotive industry and other fields. Fig. 11(b) shows the fracture strain and compressive yield strength of the three RHEAs and other typical RHEAs reported previously [5,6,15,20,21,47–56]. Compared with some typical RHEAs, the RHEAs prepared in this work have significantly higher specific yield strength and plasticity. Some RHEAs with Si addition (marked as green symbols in Fig. 11(b)), such as $HfMo_{0.5}NbTiV_{0.5}Si_{0.5}$ [5], $NbMoTaWSi_{0.25}$ [6], $HfNbSi_{0.5}TiVZr$ [20], show specific yield strength less than 210 MPa·cm^3/g.

The compressive fracture morphologies of the RHEAs are shown in Fig. 12. For the H alloy (Fig. 12(a) and (d)), obvious cleavage plane appears on the fracture surface, showing typical brittle intergranular fracture. With the addition of La and Y, the fracture surfaces become significantly rougher. Fig. 12(b) and (e) show the fracture morphologies after adding La. It can be seen that there are many small broken particles attached to the fracture surface, which displays obvious river-like patterns. It can be seen from Fig. 12(c) and (f) that after adding Y, more cleavage steps, rock sugar shaped fractures and tearing edges appear on the fracture surface. Although the addition of La and Y improves the mechanical properties of $NbMoTiVSi_{0.3}$ RHEAs, they still exhibit brittle fracture.

Conclusion

The effects of 0.5 at.% La and 0.5 at.% Y on the microstructure and mechanical properties of $NbMoTiVSi_{0.3}$ RHEAs were studied. The main conclusions are as follows:

(1) All the three RHEAs contain BCC and M_5Si_3 structure. A large amount of La and Y distribute at the phase interface of BCC and M_5Si_3, facilitating the spheroidization and refinement of M_5Si_3. That is, the shape factor K of M_5Si_3 increases from 0.26 to 0.42 for 0.5 at.% La and 0.38 for 0.5 at.% Y respectively, while the grain factor D decreases from 15.17 to 8.21 for 0.5 at.% La and 9.03 for 0.5 at.% Y respectively.

(2) With the addition of La and Y, the fraction of LAGBs increases from 18.4 % (H alloy) to 25.8 % (H-La alloy) and 23.3 % (H-Y alloy), the average KAM values increase from 0.642 (H alloy) to 0.789 (H-La alloy) and 0.874 (H-Y alloy), and the average Schmid factors of the H, H-La and H-Y alloys are 0.452, 0.488 and 0.444, respectively.

(3) A small amount of La and Y can dissolve on the BCC and M_5Si_3 phases, which facilitates the increase of H, E and H^3/E^2. For example, with the addition of Y, the H, E and H^3/E^2 of BCC phase increase from 13.03 GPa to 14.78 GPa, 218.93 GPa to 229.99 GPa, 0.047 GPa to 0.067 GPa, respectively.

(4) The yield strength, compression plasticity and specific strength of RHEAs increases from 1641.10 MPa, 12.68 %, 236.81 MPa·cm^3/g (H alloy) to 1920.93 MPa, 16.82 %, 277.59 MPa·cm^3/g for H-La alloy and 1991.59 MPa, 16.58 %, 287.80 MPa·cm^3/g for H-Y alloy, respectively.

CRediT authorship contribution statement

Tianjin Xie, Yanliang Yi, Fengshuo Jin, Jiaqing Qin, Lei Qin make the main contribution to the conception, design, data, and analyses of this work; **Yanliang Yi, Shengfeng Zhou, Shaolei Long**, make the main contribution to the acquisition, fund and analyses.

Data availability

The data sets generated and/or analyzed in this study are available from the corresponding author on reasonable request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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